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(54) **LOW-COST SOLID-STATE ALUMINUM-ION
BATTERY**

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ABSTRACT

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A method and apparatus for storing electricity using a rechargeable electrochemical cell with coiled aluminum wire anode, axially bunched carbon graphite rods as the cathode, and a solid electrolyte. The invention teaches the use of common ultra-low-cost material components, and a simple cell construction method.

Start with Cylindrical Exterior Container

(Length, l; Diameter, d)



**Place Coiled Aluminum Anodic Wire (Coiled
Length l' < l, Diameter, d'), Inside the Exterior
Container. The Negative Terminal is the Tail End of
the Coiled Aluminum**



**Place Bunched Cathodic Carbon Graphite Rods (Length
l'', Diameter, d'' for all, except Center Rod with length
l''' > l, Diameter, d''), Circles-in-Circle Arrangement,
Inside the Exterior Container . The Positive Terminal is
Graphite Rod in the Center of the cathodic Bunch**

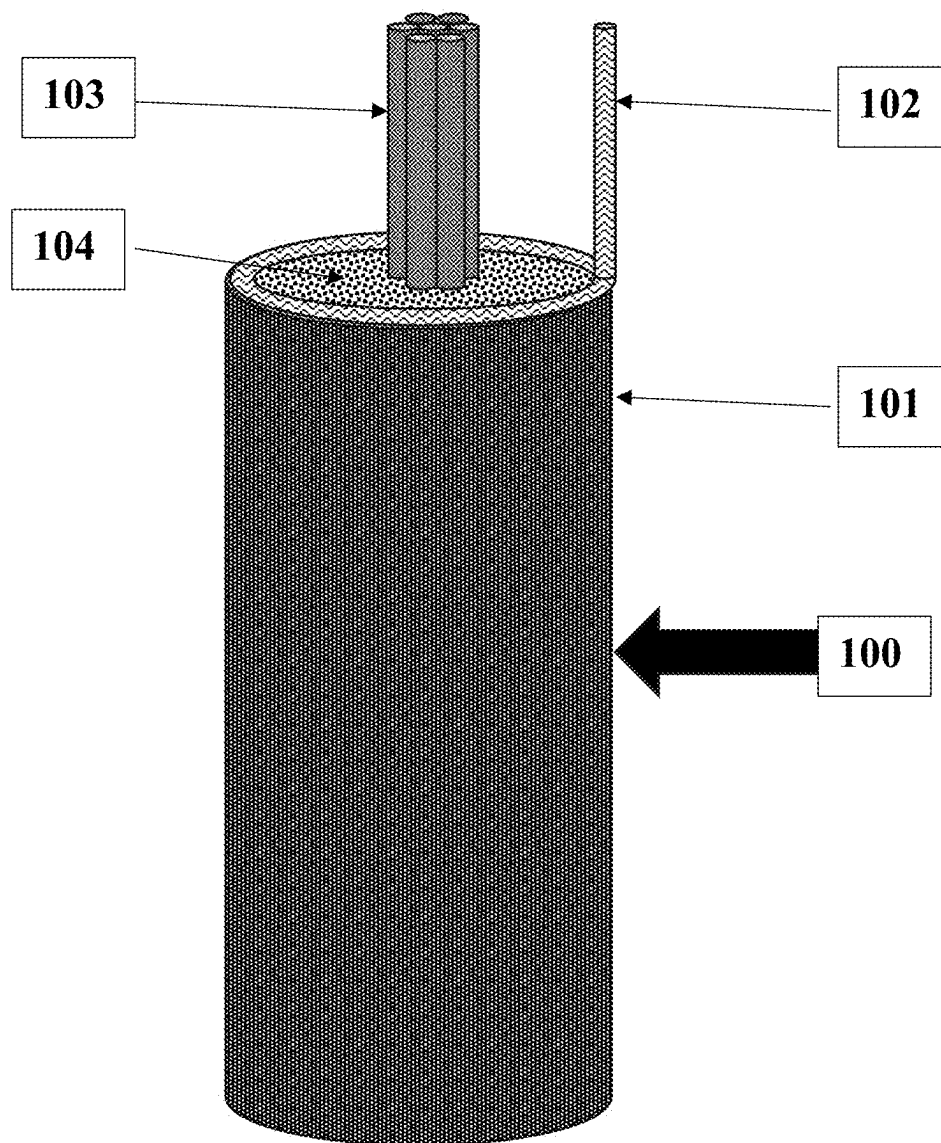
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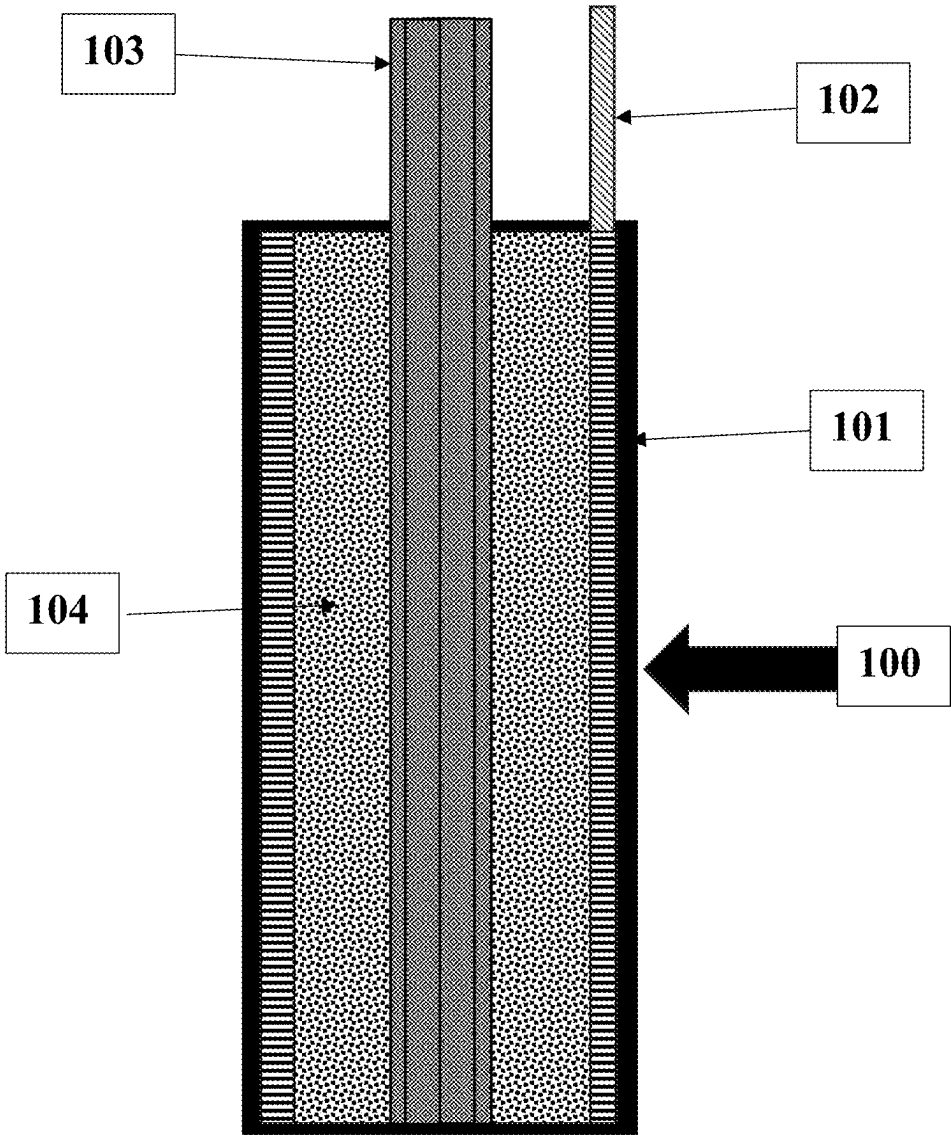


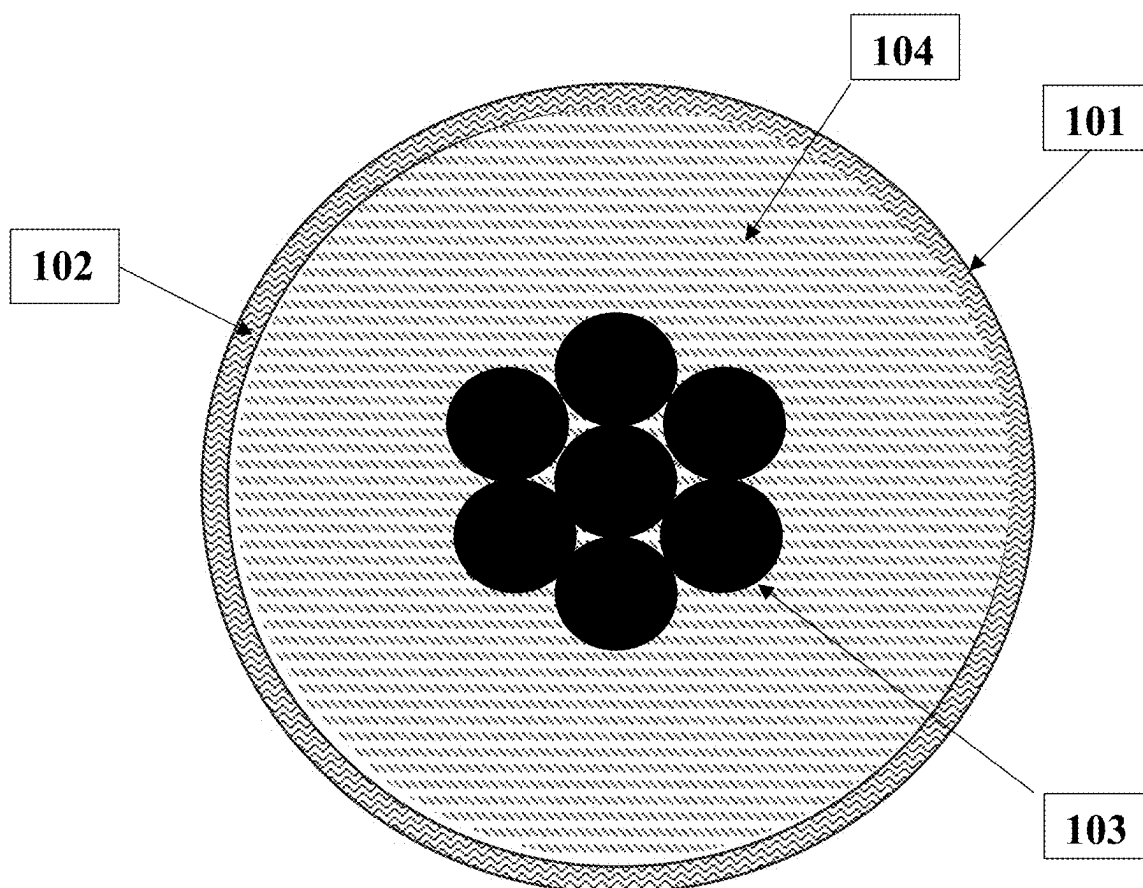
**Pour Freshly Mixed Electrolyte Into The Exterior
Container. Allow to Solidify**

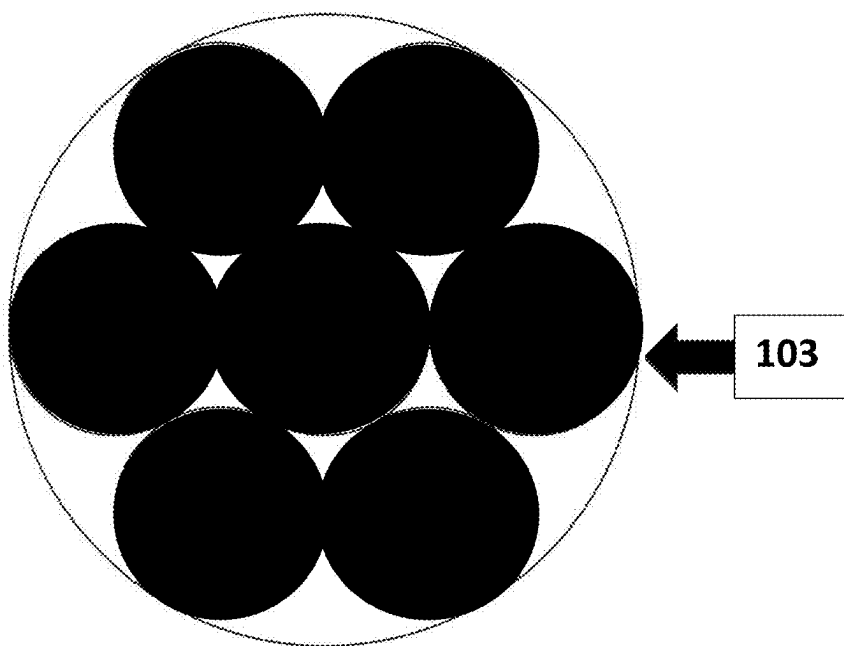


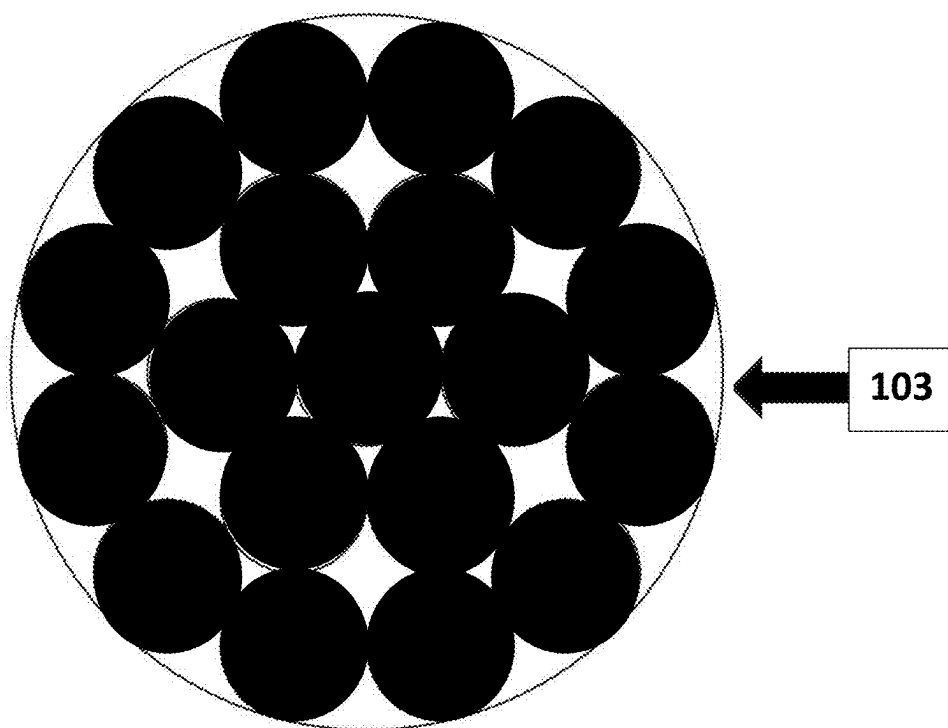
Cover and Seal External Container

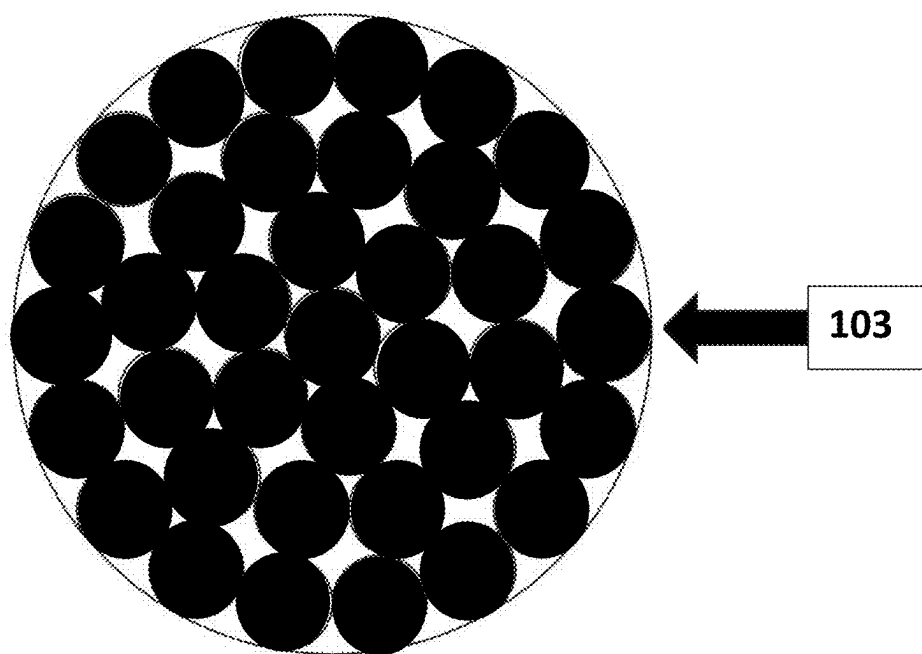


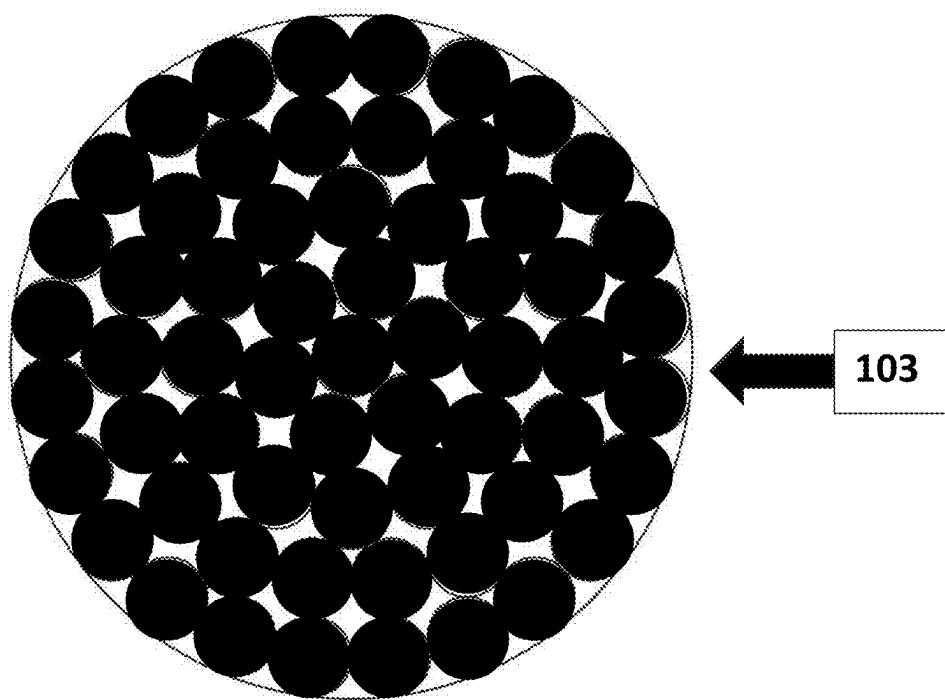












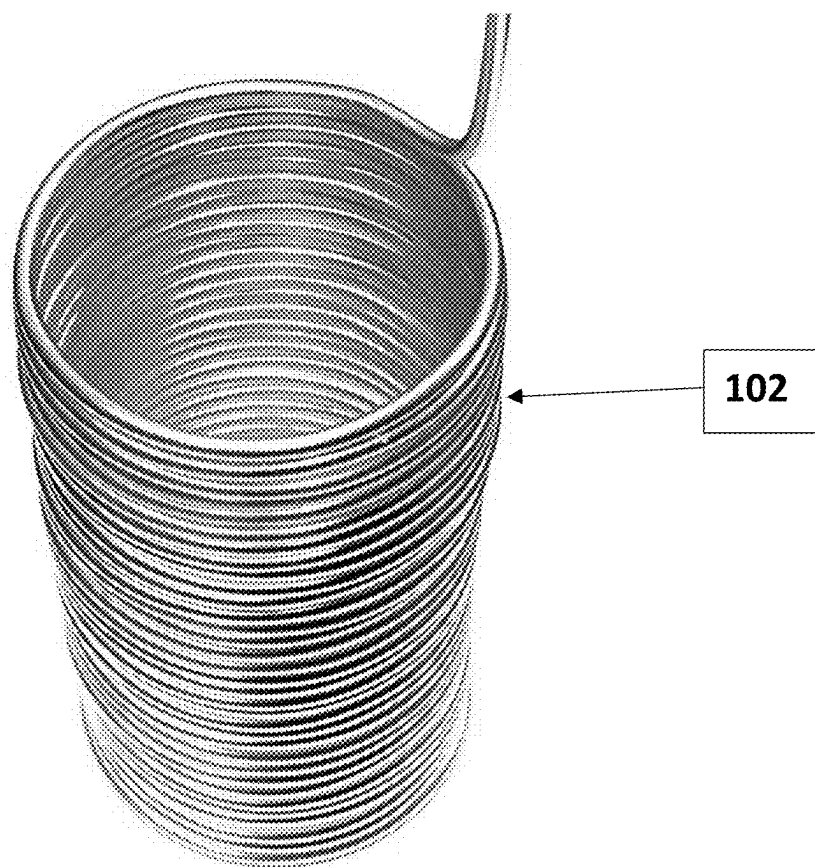


Fig. 8

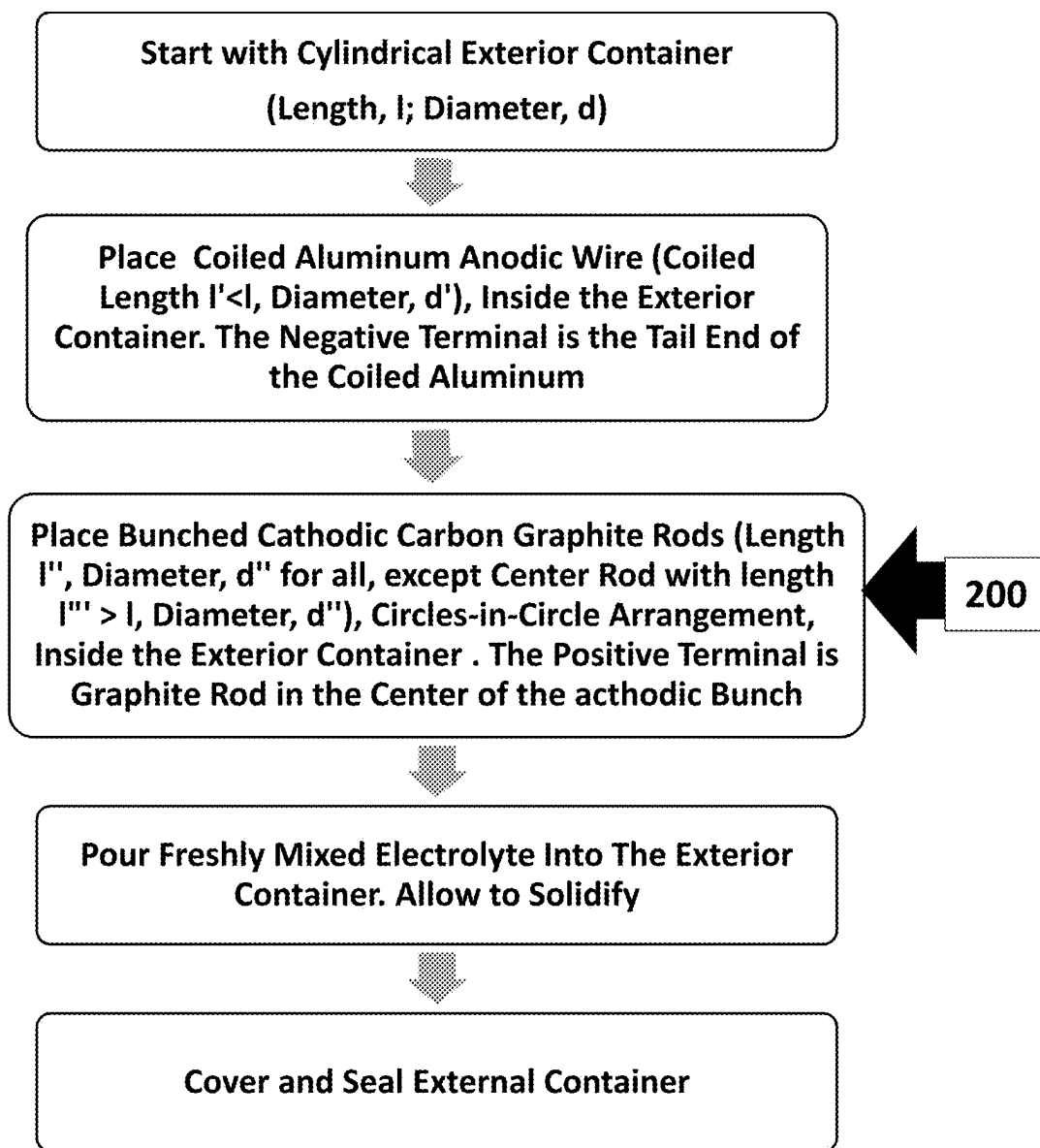


Fig. 9

LOW-COST SOLID-STATE ALUMINUM-ION BATTERY

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FIELD OF INVENTION

[0033] The present invention generally relates to electricity generation, and more particularly to a method and apparatus for electrical energy storage using rechargeable aluminum-ion electrochemical cells.

PRIOR ART

[0034] An aluminum-ion battery is a rechargeable battery in which aluminum ions serve as charge carriers. Aluminum-ion batteries consist of electrodes immersed in an electrolyte. The obvious material for the anode is aluminum, an alloy of aluminum, or a compound of aluminum. Common materials for the cathode (Pan, et al., 2022) include: carbon-based materials such as graphite, amorphous carbons, porous carbons (Li, et al., 2018). Common materials for the electrolyte include $\text{CO}(\text{NH}_2)_2$ or carbamide (also called urea), sea-salt, and acidic room temperature non-aqueous ionic liquids (IL). The ionic liquid is made of aluminum chloride (AlCl_3) and 1-ethyl-3-methylimidazolium chloride $[\text{EmIm}]\text{Cl}$ (Miguel, et al., 2020). The use of the ionic liquid as an electrolyte prevents passivation.

[0035] Aluminum-ion batteries are promising alternatives to the lithium-ion batteries commonly used in portable electronics, electric vehicles, and stationary energy for homes, commercial buildings, and grid-scale applications. Key advantages of aluminum-ion batteries include the relatively low cost, energy density, safety, and long cycle life (Leisegang, et al., 2019).

[0036] Recent investigations on aluminum-ion batteries have been conducted by:

[0037] (Wang, et al., 2017) who presented an advanced rechargeable aluminum-ion battery with a high-quality natural graphite cathode.

[0038] (Das, et al., 2017) who detailed developments and challenges faced by aluminum-ion batteries. The authors focus on the electrode materials, innovative perspectives, and future research efforts on rechargeable aluminum-ion batteries.

[0039] (Gan, et al., 2019) who designed a high-performance rechargeable aluminum battery using a graphite cathode and $\text{AlCl}_3/\text{Et}_3\text{NHCl}$ ionic liquid electrolyte.

[0040] (Levy & Ein-Eli, 2020) who discussed the potential of aluminum-ion batteries as a cost-effective energy storage technology for renewable energy sources.

[0041] (Craig, et al., 2020) who reviewed current progress in non-aqueous aluminum batteries. The authors conclude that graphite-based positive electrodes, especially cheap and abundant graphite flakes, offer the best overall performance because of the stability and high discharge potential.

[0042] (Yuan, et al., 2020) which describes the status, challenges, emerging, and outlooks of emerging rechargeable aqueous aluminum-ion battery.

[0043] (Elia, et al., 2021) who reviewed developments in different classes of aluminum-centered batteries. The focus was on “aluminum electrolyte chemistry based on “chloroaluminate melts, deep eutectic solvents, polymers, and chlorine-free formulations.”

[0044] Patents which taught implementations of the aluminum-ion batteries include:

[0045] (Archer, et al., 2014) which teaches an aluminum ion battery using an aluminum anode, a vanadium oxide material cathode, and an ionic liquid electrolyte. The vanadium oxide material cathode comprises a monocrystalline orthorhombic vanadium oxide material.

[0046] (Mukherjee & Koratkar, 2017) which describes a rechargeable battery using a solution of an aluminum salt as an electrolyte.

[0047] (Huang & Chen, 2018) which describes an aluminum-ion battery using an electrolyte comprising of an aluminum halide, a solvent and a compound made from alkyl or fluoroalkyl group.

[0048] (Gao & Chen, 2018) which discloses a method of preparing a graphene anode material by coating a graphene oxide solution on a substrate, drying, removing the substrate, performing reduction, and obtaining a graphene film with ultra-high conductivity.

[0049] (Mukherjee, et al., 2021) which describes an aqueous aluminum ion battery with improved charge storage capacity.

[0050] (Stoddart, et al., 2021) which discloses rechargeable aluminum batteries using cathodic phenanthrene-quinone unit and a graphite flake.

[0051] (Azimi & Ng, 2023) which describes aluminum-ion battery technology with an electrolyte consisting of an aluminum trichloride ($\text{Al}-\text{Cl}_3$)/trimethylamine hydrochloride ionic liquid, aluminum metal as the anode material, and a compatible cathode active material.

[0052] (Su, et al., 2020) which describes an aluminum-ion battery with a cathode comprising of a layer of recompressed exfoliated graphite or carbon material that is oriented in such a manner that the layer has a graphite edge plane in direct contact with the electrolyte and facing the separator.

[0053] (Caban-Acevedo & Yushin, 2023) which describes rechargeable batteries with alkali metal ion cathodes, aluminum metal-based anodes and displacement electrolyte.

BACKGROUND OF THE INVENTION

[0054] Aluminum-ion batteries are emerging as better alternatives to lithium-ion batteries, the leading choice for wireless devices, computers, electric mobility, and small-scale to grid-scale stationary energy storage applications. The advantages of aluminum-ion batteries over lithium-ion batteries include: cost-effectiveness of aluminum because of its abundance in the Earth's crust; safety because aluminum-ion batteries are less prone to thermal runaway and fire hazards; significantly higher theoretical energy density, 1060 Wh/kg, versus 406 Wh/kg) theoretical limit for lithium-ion batteries; high recyclability of aluminum; longevity due to the large number of charging and discharging cycles of aluminum-ion batteries.

This invention uses the advantages inherent in the electrochemistry of aluminum-ion batteries to teach the construction of an ultra-low-cost solid-state rechargeable battery. The electrodes maximize electrochemical reaction surfaces. The cells are easy to build. The cell components are affordable and widely available materials.

SUMMARY OF THE INVENTION

[0055] According to the present invention there is provided a method of storing electrical energy using a membrane-free electrochemical cell comprising a cylindrical exterior container, a coiled anodic aluminum wire, bunched carbon graphite cathodic rods, and a solid electrolyte based on a compound mixture of urea, sea-salt, and sodium silicate.

[0056] An advantage of the present invention is the provision of a method and apparatus for converting electricity into chemical energy stored in an aluminum-ion cell.

[0057] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes a solid electrolyte.

[0058] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes a coiled anodic aluminum wire.

[0059] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes bunched cathodic carbon graphite rods.

[0060] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which is membrane-free.

[0061] Still other advantages of the invention will become apparent to those skilled in the art upon reading and understanding the following detailed description, accompanying drawings and appended claims.

BRIEF DESCRIPTION OF DRAWINGS

[0062] The invention may take physical form in certain parts and arrangements of parts, a preferred embodiment and method of which will be described in detail in this specification and illustrated in the accompanying drawings which form a part hereof, and wherein:

[0063] FIG. 1 is an illustration of the isometric view of the Aluminum-Ion Cell;

[0064] FIG. 2 is an illustration of the vertical cross-section of the Aluminum-Ion Cell;

[0065] FIG. 3 is an illustration of the horizontal cross-section of the cathodic region of the Aluminum-Ion Cell;

[0066] FIG. 4 is an illustration of the horizontal cross-sectional view of the circles-in-circle arrangement of 7-rod bunched carbon graphite cathode;

[0067] FIG. 5 is an illustration of the horizontal cross-sectional view of the circles-in-circle arrangement of 19-rod bunched carbon graphite cathode;

[0068] FIG. 6 is an illustration of the horizontal cross-sectional view of the circles-in-circle arrangement of 37-rod bunched carbon graphite cathode;

[0069] FIG. 7 is an illustration of the horizontal cross-sectional view of the circles-in-circle arrangement of 61-cell bunched carbon graphite cathode;

[0070] FIG. 8 is an image of the coiled aluminum anode;

[0071] FIG. 9 is an illustration of the steps involved in the construction of the aluminum-ion cell with the coiled aluminum anode, the bunched cathodic carbon graphite rods, and the solid electrolyte.

DETAILED DESCRIPTION OF THE INVENTION

[0072] It should be appreciated that while a preferred embodiment of the present invention will be described with reference to aluminum-ion electrochemical cell, other metal-ion electrochemical cells are also suitable for use in connection with the present invention. These include zinc-ion, iron-ion, magnesium-ion, lithium-ion, calcium-ion, sodium-ion, potassium-ion, and tin-ion electrochemical cells.

[0073] In accordance with a preferred embodiment, the present invention teaches the storage of electricity using an electrochemical cell comprising a) a cylindrical exterior

container; b) a coiled anodic wire; c) a bunched group of cathodic rods; and d) an electrolyte based on a mixture of urea, sea-salt, and sodium silicate.

[0074] Referring now to the drawings wherein the showings are for the purposes of illustrating a preferred embodiment of the invention only and not for purposes of limiting same. FIG. 1 is the illustration of the isometric view of aluminum-ion electrochemical cell 100 with exterior container 101. Coiled anodic aluminum wire 102 is symmetrically placed inside exterior container 101. Bunched carbon graphite cathode 103 is at the axial center of exterior container 101. Solid electrolyte 104 occupies the entirety of the exterior container except the spaces taken up by anodic coil 102, and cathodic bunch 103.

[0075] Referring now to FIG. 2 and FIG. 8:

[0076] Anode 102. According to this invention, anode 102 is a cylindrical coil, made from spirally winding of a single aluminum wire, with diameter not exceeding 3 mm. The purity of the aluminum wire must be 99% at the minimum.

[0077] Referring now to FIG. 3:

[0078] 1. Exterior Container 101. According to this invention, exterior container 101 consists of polycarbonate rigid tubing of wall thickness not exceeding 1 mm.

[0079] 2. Cathode 103. According to the preferred embodiment of this patent, cathode 103 consists of carbon graphite rods, each of diameter not exceeding 6 mm, packed in an optimal circles-in-circle arrangement.

[0080] 3. Electrolyte 104. According to this invention, electrolyte 104 is formulated by mixing x parts by volume of powdery urea (NH_2CO) with y parts by volume of sea-salt, with z parts by volume of sodium silicate (water glass). The mixture is stirred until a uniformly smooth gel is achieved. Freshly mixed electrolyte 104, with special urea/sea-salt/water glass mix ratios, is characterized by rapid solidification. Therefore, it must be injected quickly into the cavity of electrochemical cell 100 prior to solidification.

[0081] Referring now to FIG. 4: Seven (7) carbon graphite rods 103 are bunched in optimum circles-in-circle arrangement.

[0082] Referring now to FIG. 5: Nineteen (19) electrode rods 103 are bunched in optimum circles-in-circle arrangement.

[0083] Referring now to FIG. 6: Thirty-Seven (37) electrode rods 103 are bunched in optimum circles-in-circle arrangement.

[0084] Referring now to FIG. 7: Sixty-One (61) electrode rods 103 are bunched in optimum circles-in-circle arrangement.

[0085] Referring now to FIG. 9: The steps 200 involved in the construction of aluminum-ion cell 100 are:

[0086] 1. Start with Cylindrical Exterior Container 101 (Length, l; Diameter, d)

[0087] 2. Place Coiled Aluminum Anode 102 (Coiled Length $l' < l$, Diameter, d'), Inside the Exterior Container 101. The Negative Terminal is the Tail End of the Coiled Aluminum

[0088] 3. Place Cathode 103 consisting of Bunched Carbon Graphite Rods (Length l'' , Diameter, d'' for all, except Center Rod with length $l''' > l$, Diameter, d'''), packed Circles-in-Circle Arrangement, Inside Exterior

Container **101**. The Positive Terminal is the Graphite Rod at the Center of the Cathodic Bunch **103**.

[0089] 4. Pour Freshly Mixed Electrolyte **104** into The Exterior Container **101**. Allow fresh electrolyte **104** to Solidify.

[0090] 5. Cover and Seal External Container **101**.

EXAMPLE

[0091] To establish the characteristics of the aluminum-ion electrochemical cell, according to the invention disclosed herein, charge and discharge tests were conducted on a cell of distinct size configurations, anodic coil, and cathodic carbon graphite pack. The electrolyte mixture consists of 10 ml of urea, 10 ml of sea-salt, and 120 ml of sodium silicate.

[0092] Diameter of Exterior Container: 50 mm

[0093] Length of Exterior Container: 100 mm

[0094] Anode: Cylindrical coil (Diameter: 37 mm, Height: 70 mm) made by spirally winding 2 mm (Gauge 12) aluminum wire.

[0095] Cathode: 7 graphite rods (Diameter: 6 mm; Length: 90 mm for 6 rods, and 120 mm for the 7th rod) packed circles-in-circle.

[0096] Discharge current: 10 mA

[0097] Discharge test duration: 1,285 minutes

[0098] Current Capacity: 214 mAh

[0099] Energy Produced: 305 mWh

[0100] Open Circuit Voltage: 2.5V

[0101] Average Operating Voltage: 1.4V

[0102] The present invention has been described with reference to a preferred embodiment. Obviously, modifications and alterations will occur to others upon a reading and understanding of this specification. It is intended that all such modifications and alterations be included insofar as they come within the scope of the appended claims or the equivalents thereof.

1. According to the present invention there is provided a method of storing electrical energy using a membrane-free electrochemical cell comprising a cylindrical exterior container, coiled anodic wire, bunched cathodic rods, and electrolyte based on a compound mixture of urea, sea-salt, and sodium silicate.

2. A method according to claim 1, wherein the cathode consists of bunched carbon graphite rods placed at the axial center of the exterior container.

3. A method according to claim 1, wherein the anode consists of a cylindrical aluminum coil placed near the internal periphery of the cylindrical exterior container.

4. A method according to claim 1, wherein the electrolyte is solid, consisting of a compound mixture of urea, sea-salt, and sodium silicate.

5. A method according to claim 1, wherein the electrochemical cell has no physical membrane separator.

* * * * *